

CS 601: CS Research Methodology Seminar

Special Topic 1: PhD Graduate Handbook

Adnan Siraj Rakin

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What is this handbook for?

- Provide an overview of the requirements and policies for successfully completing a computer science Ph.D. degree at Binghamton University.
- PhD Students generally need to follow the requirements and policies. Exceptions to accommodate special circumstances must be approved by the student's advisor and the Director of Graduate Studies (currently Prof. Leslie Lander). Exceptions must be documented and included in the student's academic file.
- The handbook is intended primarily for students who have already been admitted to the Ph.D. Program.



Major Milestones

Milestone	Description	Recommended Timeline (latest by)
Part A: Learning Contract	Complete Part A in consultation with the PhD advisor and Guidance Committee.	End of Month 1. Update as needed during PhD if there are changes.
Part B: PhD Qualifying Requirement	Complete PhD Qualifying Requirement.	End of Year 2
Part C: Baseline literature list for RPE	Decide with the PhD advisor on a baseline literature list for the RPE exam.	End of Year 2
Part C: RPE Approval	Submission and acceptance of the Research Proficiency Examination (RPE) report and public oral presentation.	End of Year 3
Part D: Prospectus	Submission and acceptance of the Prospectus report and a public oral presentation.	End of Year 3.5
Part E: Publications and presentations	Publication of student's research results in conferences and/or journals and public presentations to meet "Teaching Requirements"	Update whenever you have new peer-reviewed publications and/or public presentations.
Part F: Dissertation	Submission and acceptance of the Dissertation and a public oral defense.	End of Year 6

Learning Contract

- The Learning Contract records various requirements and milestones of a PhD student towards a PhD degree, in consultation and agreement with the student's Advisor and/or Guidance Committee.
- This Learning Contract must follow all the requirements of the CS PhD Program.
- This Learning Contract consists of six parts and will be completed gradually:
 - Part A: Basic information and course selections
 - Part B: about PhD Qualifying Exam
 - Part C: about the Research Proficiency Exam (RPE)
 - Part D: about PhD Prospectus
 - Part E: List of publications
 - Part F: about for PhD dissertation defense

Learning Contract (continued)

Completing the Learning Contract:

- Join the [Google Classroom for PhD Status Tracking](#)
- Submit Part A by the end of month 1 in consultation with your PhD advisor.
- Submit Part B as soon as your PhD Qualifying requirements are met.
- Submit Part C along with your report as soon as you pass RPE.
- Submit Part D along with your report as soon as you pass Prospectus.
- Update Part E whenever you publish a new paper in a journal, workshop, or conference.
- Follow Part F instructions before and after you defend your dissertation.

Full-time status requirement

- The Graduate School requires Ph.D. students receiving tuition scholarships to register for at least 9 credits each semester. CS 697 (Advanced Independent Study), CS 698 (pre-dissertation research), CS 699 (dissertation) can be used to satisfy this requirement.
- Students who have been admitted to PhD candidacy and have completed all courses can apply for the **course-complete status**. Students in course-complete status generally cannot take regular courses (can take CS 699) and can satisfy the full-time status by taking just one credit of CS 699, which leads to lower tuition and fees.
- Only full-time students are eligible for financial assistance from School of Computing (SoC).

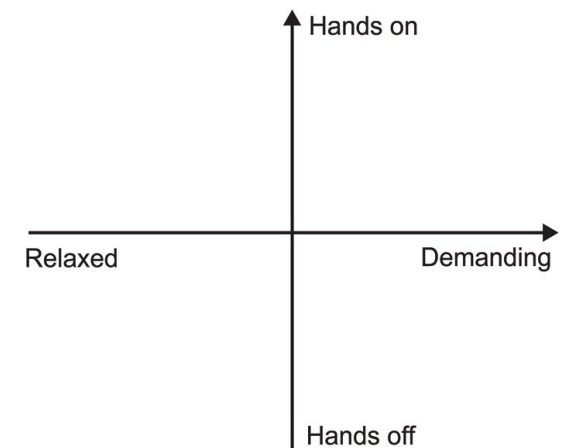
Advisor Selection

- Most PhD students should already have a faculty advisor when starting their PhD program.
- If you don't have one yet, please talk to Prof. Leslie Lander (Graduate Director) or Prof. Ping Yang (Associate Director for Research and Graduate Programs) about how to select one.
- Your advisor will be the most important person during your PhD study and possibly in your future career.
- The advisor will help the student form the Ph.D. Guidance Committee, select a research topic, evaluate progress, and guide the student toward the completion of their Ph.D. dissertation.

Advisor Selection (continued)

- If you have a chance to select an advisor, what factors should you consider?
- For many, scholarship/financial aid is the most important factor.
- Other significant factors include
 - The prospective advisor's research areas and interests.
 - The prospective advisor's research approach – theoretical, empirical, systems, applications, ...
 - The prospective advisor's recent track record – publication, funding, where students get their first employment, ...
 - The prospective advisor's supervisory philosophy and style
 - The prospective advisor's status or whether is tenured.
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- Knowing the styles of your advisor can help you work with your advisor better.
- Remember: it's not just you who do the selection, your prospective advisor too.

Choose a PhD Supervisor



From Ling & Yang Book

What if you cannot work well with your advisor?

- There could be many reasons for this to happen:
 - Student not making good progress
 - Student not sufficiently independent
 - Advisor not giving good advice (topic, research method, ...)
 - Advisor not using student appropriately, e.g., student asked to work on un-related project, student's other needs (vacation, family, ...) not adequately considered
 - Unfriendly working environment
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- Talk to Professor Ping Yang about such problems and ask for her advice.
- Sometimes, changing advisor could be the best option.

Termination of PhD Study

- There could be different circumstances that lead to the termination of a student's PhD study:
 - Student quits the PhD program for personal reasons.
 - Student fails to pass the qualifying exams.
 - Student has low GPA.
 - Student no longer has funding.
 - Academic dishonesty issues.
 - Supervisor no longer wants to supervise the student, usually for not making adequate progress in research or having irreconcilable differences. Same result with a different supervisor.
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How PhD students are funded?

- Almost all full-time PhD students are funded mainly by one of the following sources:
 - **Research Assistantship** funded by the advisor's research grant.
 - Advisor needs to report research progress such as publications on specific research problems to the funding agency and student(s) funded on the grant have the obligation to work on the problems diligently and make good progress.
 - The research project from the grant and the student's PhD research project don't have to be the same but would be better if they are the same.
 - **Graduate Assistantship** funded by the advisor's startup fund.
 - Advisor needs to make progress on new research problems that help them write grant proposals to fund their PhD students.
 - It is also in the student's best interest to work hard and make good progress in the research.
 - **Teaching Assistantship** funded by the School of Computing to support a specific faculty's research.
 - Students with TA funding should focus on their PhD research while fulfilling their TA responsibilities.
 - **Special government programs/fellowships**

Form the PhD Guidance Committee

- PhD students, with the help of their advisor, must form their Ph.D. Guidance Committee within the first year of their PhD study.
- The committee must consist of the advisor, and at least two additional members of the faculty from the School of Computing in Binghamton University.
- Qualified faculty from outside the School of Computing or Binghamton University and qualified researchers from industrial research labs could be added to the Guidance Committee subject to approval.
- Committee members should have expertise that are related to some aspects of the student's dissertation research.
- Guidance Committee can change over the course of the student's PhD study.

Residence Requirement

- Students who did not receive an M.S. degree from Binghamton University are required to have a minimum of 24 credit hours in residence in order to graduate.
- CS 698 and CS 699 do not count toward the residence requirement.
- Students who earned their master's degree from the School of Computing at Binghamton University within the previous five years are deemed to have already fulfilled the residence requirement.
- Residence requirement is one of the requirements for admission to PhD candidacy and students should try to satisfy this requirement within the first three semesters.

Course Requirements with a prior MS degree

- Must complete a minimum of 30 credits beyond the M.S. degree. Among the 30 credits
 - At least 18 credits (6 courses) need to come from courses not including CS 597 (Independent Study) or CS 697 (Advanced Independent Study). Among the 6 courses
 - students must take CS 601 and at least one other 3-credit 600-level course specified in the student's Learning Contract
 - students who did not receive their MS degree from Binghamton University's SoC must take **at least two** of the **core courses** (CS 551 Systems Programming, CS 571 Programming Languages, CS 575 Design and Analysis of Computer Algorithms) and the rest from 500-level graded courses closely related to the student's research as specified in the Learning Contract
 - students who have completed an MS CS at Binghamton University can count their CS courses already taken during MS CS towards this PhD core course requirement, subject to the above restrictions. However such students must still satisfy the 30 credits of "Credit and Coursework Requirements" for PhD students mentioned above using other CS courses not yet taken.
 - At most 6 credits may come from CS 597 and CS 697.
 - The remaining credits may come from Pre-dissertation Research (CS 698) and Dissertation Research (CS 699).
- To graduate, GPA must be at least 3.0 overall in graduate courses and also the courses taken to satisfy the M.S. or Ph.D. degree requirement.

Course Requirements without a prior MS degree

- Must complete a minimum of 60 credits beyond the B.S. degree. Among the 60 credits
 - At least 42 credits (14 courses) need to come from courses not including CS 597 or CS 697. Among these 14 courses
 - students must take CS 601 and at least one other 3-credit 600-level course specified in the student's Learning Contract
 - students must take all the **core courses** (CS 551 Systems Programming, CS 571 Programming Languages, CS 575 Design and Analysis of Computer Algorithms) plus taking another graded course closely related to the student's research as specified in the Learning Contract
 - the remaining courses are from other 500-level or 600-level electives
 - At most 12 credits may come from CS 597 and CS 697.
 - The remaining credits may come from CS 698 and CS 699.
- We rarely admit students with only a BS degree directly into our PhD program.

Ph.D. Qualifying Examination

- Every Ph.D. student must pass the *Ph.D. Qualifying Exam* (also known as the *Ph.D. Comprehensive Examination*). The purpose of this exam is to ensure that every Ph.D. candidate has the breadth of general computer science knowledge expected of graduate students.
- The Qualifying Examination should be completed within the first three semesters.
- This exam covers the following four required core areas:
 - At least two courses from CS 551, CS 571 and CS 575
 - The remaining courses from 500-level electives in the candidate's research area – the course(s) should be listed in the student's Learning Contract Part A.
- The Qualifying Exam will be waived if the student receives at least a grade of B+ for every core course with an average grade of at least 3.5.
- If a student's Qualifying Exam is not waived, the student must pass the Qualifying Exam in any core course where the student receives a grade of B+ or lower.

Research Proficiency Examination (RPE)

- The purpose of RPE is to determine the breadth and depth of the student's preparation to undertake significant original research in a specific research area.
 - Breadth refers to the broad knowledge of prior related research in the area of investigation, the pros & cons of each prior work, and systematic classification of existing research results.
 - Depth refers to expertise in the detailed technical subject matter in the area, identification of open research problems, and conveying the importance of solving these open problems.
- The RPE will consist of two components:
 - A written RPE report prepared by the student.
 - It should be at least 10 pages (single-spaced, 12-point font, 1-inch margins), summarizes each study reviewed, and outlines future research directions.
 - A peer-reviewed full-length (at least 8 pages) first-author paper accepted by a quality conference or a journal could be used as a substitute.
 - An oral RPE presentation to the student's Guidance Committee, which is open to others. There is an Q&A session after the presentation.
- Students who fail RPE twice will be dismissed from the PhD program.

Admission to PhD Candidacy

- To be eligible for admission to PhD Candidacy (or ABD – All But Dissertation status), a student must satisfy the following conditions:
 1. Completed the course requirement (24 credits of courses, not counting CS 698 and CS 699 credits, if the student has a prior M.S. degree; 54 credits of courses otherwise).
 2. Passed the Qualifying Exam
 3. Passed the RPE.
- Once eligible, the Guidance Committee will complete the form *Recommendation for Admission to Candidacy for Doctoral Degree* for the signature of the Director of Graduate Studies. The status of candidacy will be officially conferred by the Dean of the Graduate School upon recommendation of the SoC.
- Students should aim to achieve the ABD status as soon as possible.

Teaching Requirement

- All PhD students are required to fulfill the teaching requirement before graduating, which is designed to ensure that all Ph.D. graduates have the basic skills to make effective technical presentations to an audience.
- A student can fulfill this requirement by completing one of the following:
 - Teach a CS course (e.g., a summer course) at Binghamton University as an instructor of record.
 - Make at least 6 public presentations with at least one of them being a colloquium in the CS School of Computing colloquium series. Acceptable public presentations include:
 - RPE presentation
 - Prospectus presentation
 - Dissertation defense presentation
 - Any public presentation that is announced at least one week in advance
 - Lecture (including guest lecture) given to students in a normal class
 - Paper presentation at a technical conference/symposium/workshop

Watson Future Faculty Fellows (W3F) Program

Beyond the Degree: Your Academic Career Starts Here!

The **Watson Future Faculty Fellows (WF3) Program** aims to facilitate professional training of selected Watson PhD students for successful faculty careers in academia. Fellows will receive specialized training and practical experience in higher-education teaching, proposal writing, manuscript review, and other essential skills in academic settings.

- ✓ **Ideal professional training for future faculty** – shadowing faculty & teaching a course
- ✓ **Additional training sessions** – proposal writing, manuscript review, teaching, etc.
- ✓ **A \$5,000 summer/winter award upon completion of the program**
- ✓ **Special recognition that boosts your credentials and competitiveness in the academia job market**

❖ Eligibility

- ✓ Enrolled in a Watson PhD program with good academic standing
- ✓ Has the intent to become a faculty member in a research university in the future career
- ✓ Nominated by the Department/School

❖ Program Structure

- ✓ One semester (or session) of shadowing a faculty member in teaching a course
- ✓ One semester (or session) of teaching the same course as the instructor
- ✓ Additional training sessions on proposal writing, manuscript review, teaching effectiveness, etc.
- ✓ Recognition ceremony at the completion

❖ How to apply

- ✓ Students cannot apply directly. Interested students should contact their Department Chair/School Director and the Graduate Program Director to be nominated for the WF3 program.



- Completing this program automatically fulfill the PhD Teaching Requirement.
- I recommend that you consider this program after admission to PhD candidacy.
- Interested students should discuss with their advisor for guidance and support first. Then talk to Prof. Ping Yang for approval and the process.

Poll: Who are interested in the program?

Prospectus

- The primary purpose of the Prospectus is to assess the student's progress towards completing the Ph.D. dissertation and specific plans for achieving the research objectives.
- It is recommended to use the PhD dissertation formatting guide to prepare the prospectus report (see [grad-school/academic-support/submit-thesis-dissertation_I](#)). This will significantly reduce the effort in writing the dissertation later.
- The student will give an oral presentation of the prospectus to the Guidance Committee and will be open to the public. An announcement about the prospectus presentation must be made in the School of Computing at least one week before the presentation.
- The presentation will be followed by a closed session where the committee members are free to question the student on any topics that they feel are in any way relevant to the student's objectives and career preparation. The student is expected to show complete familiarity with the current and past literature of his/her research area.

Prospectus (continued)

- The major requirements of the Prospectus are:
 1. **Background:** The student must be thoroughly familiar with the background and current status of the intended research area.
 2. **Justification:** The student must justify the effort to be expended in the research in terms of the value of the results expected, and the challenge of that research as evidence of research competence at the Ph.D. level.
 3. **Research Plans:** The student must describe well-defined research objectives and concrete plans for achieving the research objectives before completion of Ph.D.
 4. **Evidence of progress:** Most importantly, the student must offer evidence of progress in achieving these objectives. The evidence could be in terms of preliminary research results, prototype software/systems developed and/or publications in which the student has played a significant role. In particular, publications from the student at well-known venues at this stage of the student's Ph.D. study can be considered to be strong evidence of satisfactory progress.

Dissertation and Dissertation Defense

- The dissertation and dissertation defense should conform to all regulations found in the Graduate School Manual ([grad-school/resources/policies-procedures/manual/academic-doctoral](#)).
- Prior to the dissertation defense, an Outside Examiner needs to be selected and added to the Guidance Committee at least one month in advance. The Graduate School must approve the external examiner since they represent the Graduate School at the defense. An Outside Examiner may be selected from the pre-approved list provided by the Graduate School ([grad-school/resources/policies-procedures/manual 5](#)).
- A draft of the dissertation must be prepared by the student and shared with the Guidance Committee 2-3 weeks before the defense.
- The defense must be announced to the SoC at least 10 days before the defense.
- Upon successful completion of the dissertation defense and acceptance of the written dissertation by the entire Guidance Committee, the electronic *Recommendation for Award* form will be approved by each member and the Graduate Director.

Dissertation Defense

- A typical dissertation defense process:
 1. The chair of the Guidance Committee (usually the student's advisor) introduces the candidate.
 2. The student does defense presentation. Guidance Committee members and the audience could ask questions during the presentation.
 3. After the presentation is done, everyone in the audience, including committee members, is given chance to ask questions.
 4. After questions from the audience have been answered, the audience leaves the defense. The committee members ask additional questions in a closed session.
 5. After the committee members asked their questions, the candidate leaves the room. The committee has a closed session to discuss what changes the student must make to finalize the dissertation and whether to approve it.
 6. The student returns to the room. The committee chair provides a summary of the committee's discussions to the student. Then the defense is over.
- Student follows up with suggested changes from the Guidance Committee and submit the final dissertation.

Five-Year Limits

- Ph.D. students must be admitted to candidacy within five years of entering the doctoral program.
- They must complete all remaining degree requirements within five years after admission to candidacy.
 - Approved leaves of absence do not count toward the time limit.
- Students may request an extension of the time limits by submitting a petition to the Graduate Director for approval by the Graduate Director and the Graduate School. The petition form is available at [extension of five-year link f.](#)